

NumPy Interview Questions with Answers

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1. What is NumPy?

Answer:

NumPy is a Python library used for numerical and mathematical operations.

It provides:

- Arrays
- Mathematical functions
- Matrix operations
- Fast computation

2. Why is NumPy faster than Python lists?

Answer:

- They use less memory
- Data type is fixed
- Operations are implemented in C language
- Vectorized operations increase speed

3. How to import NumPy?

Answer:

```
import numpy as np
```

4. What is a NumPy array?

Answer:

A NumPy array is a collection of elements of the same data type stored in contiguous memory locations.

Example:

```
import numpy as np
```

```
arr = np.array([1,2,3,4])
```

```
print(arr)
```

Output:

```
[1 2 3 4]
```

5. Difference between List and NumPy Array?

Python List:

- Slower
- More memory
- Different data types
- Limited math operations

NumPy Array:

- Faster
- Less memory
- Same data type
- Powerful math operations

6. How to create zeros and ones arrays?

Answer:

```
import numpy as np
```

```
print(np.zeros((2,2)))
```

```
print(np.ones((3,3)))
```

7. What is array slicing?

Answer:

Slicing is used to access specific parts of an array.

Example:

```
arr = np.array([1,2,3,4,5])
```

```
print(arr[1:4])
```

Output:

```
[2 3 4]
```

8. What is reshaping in NumPy?

Answer:

Reshape changes the dimensions of an array without changing data.

Example:

```
arr = np.array([1,2,3,4,5,6])
```

```
print(arr.reshape(2,3))
```

Output:

```
[[1 2 3]
```

```
[4 5 6]]
```

9. What is broadcasting in NumPy?

Answer:

Broadcasting allows NumPy to perform operations on arrays with different shapes.

Example:

```
arr = np.array([1,2,3])
```

```
print(arr + 5)
```

Output:

```
[6 7 8]
```

10. What is the difference between reshape() and resize()?

reshape():

- Returns new array
- Original data unchanged

resize():

- Changes original array
- Original modified

11. How to find array dimensions?

Answer:

```
arr = np.array([[1,2],[3,4]])
```

```
print(arr.ndim)
```

Output:

2

12. How to find shape of array?

Answer:

```
arr = np.array([[1,2,3],[4,5,6]])
```

```
print(arr.shape)
```

Output:

(2, 3)

13. How to generate random numbers in NumPy?

Answer:

```
import numpy as np
```

```
print(np.random.rand(3))
```

14. What is vectorization?

Answer:

Vectorization means performing operations on entire arrays without loops.

Example:

```
arr = np.array([1,2,3])
```

```
print(arr * 2)
```

Output:

```
[2 4 6]
```

15. How to calculate mean in NumPy?

Answer:

```
arr = np.array([1,2,3,4])
```

```
print(np.mean(arr))
```

Output:

```
2.5
```

16. How to find maximum and minimum values?

Answer:

```
arr = np.array([10,20,5,40])
```

```
print(np.max(arr))
```

```
print(np.min(arr))
```

Output:

```
40
```

```
5
```

17. What is flatten() in NumPy?

Answer:

flatten() converts multidimensional arrays into 1D arrays.

Example:

```
arr = np.array([[1,2],[3,4]])
```

```
print(arr.flatten())
```

Output:

```
[1 2 3 4]
```

18. Difference between copy() and view()?

copy():

- Separate data
- Changes not reflected

view():

- Shares original data
- Changes reflected

19. How to concatenate arrays?

Answer:

```
a = np.array([1,2])
```

```
b = np.array([3,4])
```

```
print(np.concatenate((a,b)))
```

Output:

```
[1 2 3 4]
```

20. What are NumPy advantages?

Answer:

- Fast computation
- Efficient memory usage

- Powerful mathematical functions
- Multi-dimensional arrays
- Used in AI, ML, and Data Science